

AYUSH KULAL

Backend & Applied AI Engineer

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SUMMARY

Backend and Applied AI Engineer experienced in building production AI agents, REST APIs, enterprise automation systems, and intelligent developer tools using Python, Java, Spring Boot, FastAPI, and PostgreSQL. Skilled in MCP, agentic AI, RAG, performance engineering, authentication, and scalable backend systems, with hands-on experience shipping production software in cross-functional engineering teams.

EXPERIENCE

AI Agent Developer (Internship) – SpurQLabs

Jun 2026 – Present

- Designed and developed **3 production-ready AI agents** for enterprise automation, streamlining software testing workflows.
- Built scalable backend services and intelligent automation systems performance, reliability, and maintainability
- Designed and delivered AI automation solutions across QA, Sales, and Marketing workflows using MCP-based architectures.
- Participated in software architecture, technical design, debugging, and iterative product enhancements across multiple AI initiatives.

Product Lead & Engineer (Self-Employed) - [Anchorlit](#)

Jul 2024 – Jun 2026

- Building AI-powered and full-stack web applications with scalable backend systems and modern UI/UX
- Leading product development, API architecture, deployment, and automation workflow implementation
- Managed the complete software development lifecycle from requirements gathering and architecture to deployment and maintenance.

PROJECTS

QAForge - Autonomous AI QA Platform | [Live](#) | [GitHub](#)

Python · FastAPI · Next.js · Supabase · LangGraph · Groq · Webhooks

Autonomous AI-powered QA platform for repository analysis, runtime testing, and webhook-triggered audits.

- Built a multi-agent AI audit orchestration pipeline with live execution logs, findings generation, and AI-generated reports
- Implemented GitHub OAuth integration with dynamic repository fetching and webhook-based autonomous audit triggering
- Designed a real-time operational dashboard with streaming pipeline stages, audit synchronization, and responsive UI
- Deployed frontend on Vercel and backend on Render with Supabase persistence and production-ready environment management

Ragora - Agentic RAG Platform | [Live](#) | [GitHub](#)

Python · FastAPI · LangChain · Vector DB · LLM APIs · PostgreSQL · Redis · Docker

Multi-tenant agentic RAG platform for secure AI chatbot deployment and intelligent retrieval workflows.

- Implemented hybrid retrieval using vector similarity, full-text search, and Reciprocal Rank Fusion (RRF) for improved retrieval accuracy
- Designed strict multi-tenant isolation at database and API layers to prevent cross-tenant data exposure
- Built conversation-aware query rewriting using LLM context and chat history for improved follow-up handling
- Added confidence-gated response generation with fallback and escalation workflows to reduce hallucinations

Bloom Events - Event Coordination Platform | [Live](#) | [GitHub](#)

Spring Boot · React · PostgreSQL · Supabase · OAuth 2.0 · JWT

Multi-step event booking platform with Google OAuth login and dynamic cost calculation

- Built a full-stack event booking platform with Google OAuth login and secure user authentication flow across modules
- Implemented REST APIs using Spring Boot and JPA for managing events, services, bookings, and relational data efficiently
- Built dynamic pricing logic to calculate real-time costs based on selected services and user inputs
- Deployed frontend on Vercel and backend on Railway with proper environment configuration and seamless API integration

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, SQL

Backend: Spring Boot, Spring Security, REST APIs, FastAPI, JPA, Hibernate

Frontend: React, Tailwind CSS, Next.js, HTML, CSS, TypeScript

Database: PostgreSQL, MySQL, SQLite, MongoDB, Redis, Vector Databases

AI / ML: MCP, Agentic AI, RAG, LangChain, LangGraph, LLM Integration, Semantic Search, Prompt Engineering

DevOps: Docker, Git, GitHub, AWS, Vercel, Render, Railway

Tools & Technologies: Postman, Maven, Webhooks, OAuth 2.0, JWT

EDUCATION

B.E. Computer Engineering - AISSMS College of Engineering, Pune

2024 – 2027 | CGPA: 8.2

Diploma in Information Technology - AISSMS Polytechnic, Pune

2021 – 2024 | 88.88%

ACHIEVEMENTS

- Received Letter of Recommendation (LOR) from **Pune Municipal Corporation (PMC)** for “[EcoWatchAI](#)” project
- Secured 1st rank among 240+ participants in State-Level TechCode War 2023 competition